

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)			electron promoted to a higher energy level (1)	2			2		
				falls back down to a lower energy level (1)						
	(b)	(i)		each series corresponds to electrons falling to different energy levels	1			1		
		(ii)		energy difference between the shells decreases / energy levels get closer together	1			1		
	(c)	(i)		ionisation of the atom / point at which electron leaves the atom	1			1		
		(ii)		$\Delta E = hf$ and $f = c/\lambda$ (1) $\frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{92.1 \times 10^{-9}} \quad (1)$ $2.18 \times 10^{-18} \text{ J} \quad (1)$ $1312 \text{ kJ mol}^{-1} \quad (1)$	1					
						3		4	4	

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	(d)	(i)		$n(\text{Al}) = 4.85 \times 10^{-3}$ (1) $n(\text{H}_2) = 7.28 \times 10^{-3}$ (1) volume $\text{H}_2 = 178 \text{ cm}^3$ (1)		3		3	2	
		(ii)		$V = \frac{1 \times 178 \times 323}{298 \times 1.6} (1)$ $V = 121 \text{ cm}^3$ (1) if 200 cm^3 used award (2) for $V = 135 \text{ cm}^3$ Credit alternative method using $pV = nRT$ $V = \frac{7.28 \times 10^{-3} \times 8.31 \times 323}{161600} (1)$ $V = 121 \text{ cm}^3$ (1)		2		2	2	
				Question 10 total	6	8	0	14	8	0